

Conditional deletion of the TGF- β type II receptor in Col2a expressing cells results in defects in the axial skeleton without alterations in chondrocyte differentiation or embryonic development of long bones

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Received for publication 7 February 2004, revised 1 June 2004, accepted 3 August 2004

Available online 11 September 2004

Abstract

Members of the TGF- β superfamily are secreted signaling proteins that regulate many aspects of development including growth and differentiation in skeletal tissue. There are three isoforms of TGF- β that act through the same heteromeric receptor complex. To address the question of the role of TGF- β signaling in skeletal development, we generated mice with a conditional deletion of the TGF- β type II receptor gene (*Tgfb2*) specifically in Col2a expressing cells using the Cre/lox recombinase system. Alizarin red-/Alcian blue-stained skeletons were prepared from embryos at 17.5, 15.5, and 13.5 days of gestation. Col2acre^{+/−}; *Tgfb2*^{loxP/loxP} and Col2acre^{−/−}; *Tgfb2*^{+/loxP} skeletons were compared. Multiple defects were observed in the base of the skull and in the vertebrae. Specifically, the size and spacing of the vertebrae were altered, and defects were detected in the closure of the neural arches. In addition, alterations in transverse processes, costal joints, and zygapophyses were detected. While the vertebral bodies were only moderately affected, the intervertebral discs (IVDs) were either missing or incomplete. Alterations in the vertebrae could be detected as early as E13.5 days. Surprisingly, alterations in length and mineralization of long bones were not detected at E17.5 days. In addition, the expression patterns of markers for chondrocyte differentiation were not altered in vertebrae or long bones suggesting that loss of responsiveness to TGF- β in chondrocytes does not affect embryonic endochondral bone formation. In contrast, mice that survived postnatally demonstrated alterations in the length of specific bones. Skeletons from Col2acre^{+/−}; *Tgfb2*^{loxP/loxP} mice were compared to those from mice null for the TGF- β 2 ligand. The differences observed between these models allow distinctions to be made between the roles of the various isoforms of TGF- β and the signaling in specific cell types. The data provide information regarding mechanisms of skeletal development and suggest that TGF- β signaling is a critical component.

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Keywords: Endochondral bone; Perichondrium; Vertebrae; Intervertebral disc; Cre/Lox

Introduction

Members of the transforming growth factor-beta (TGF- β) superfamily are secreted signaling molecules that regulate many aspects of cell growth, differentiation, chemotaxis, and connective tissue and extracellular matrix production (Chang et al., 2002; Massague et al., 2000; Roberts and Sporn,

1990). The family includes three TGF- β isoforms (TGF- β 1, 2, 3), the activins and inhibins, growth and differentiation factors (Gdfs), and the bone morphogenetic proteins (Bmps). TGF- β signals through a complex of type I and II serine/threonine kinase receptors (T β RI and T β RII). A family of type I and II receptors that mediate signaling by members of the TGF- β superfamily has been cloned (Derynck, 1994; Shi and Massague, 2003; Ten Dijke et al., 1996). Biochemical and genetic evidence argues that both type I and II receptors are required to generate a response to TGF- β (Laiho et al., 1990; Ruberte et al., 1995; Wrana et al., 1992). The current model is that TGF- β binds to T β RII on the cell surface

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(Wrana et al., 1994). T β RII is then able to recruit T β RI to form a heterotetrameric complex of two type I and two type II receptors. T β RII is a constitutively active kinase that phosphorylates the GS domain of T β RI, activating the serine/threonine kinase. Downstream targets of T β RI then transduce the signal to the nucleus. The Smad family of proteins has been identified as important transducers of TGF- β signaling (Derynck and Zhang, 1996; Shi and Massague, 2003). The current model is that Smads are phosphorylated by T β RI, translocate to the nucleus, and act as transcription factors.

The vertebral column develops from somites, segmental units of the paraxial mesoderm (Christ et al., 2000). In response to signals from the notochord and floor plate of the neural tube (Watanabe et al., 1998), the maturing somites will undergo a dorsal–ventral compartmentalization establishing the dermamyotome and sclerotome, respectively. The sclerotome will establish the vertebral anlagen through the orchestration of its dorsal, lateral, and ventral populations. The ventral sclerotome gives rise to the vertebral bodies and intervertebral discs (IVDs). The lateral sclerotome gives rise to the laminae and pedicles of the neural arches and to the ribs. Cells that migrate from the dorsomedial angle of the sclerotome into the space between the roof plate of the neural tube and the dermis form the dorsal portions of the neural arch including the spinous process. Subsequently, the cells undergo endochondral ossification to form the adult vertebrae. Endochondral ossification is the predominant means by which the majority of skeletal elements are formed. Endochondral bone formation is the process by which a cartilage template preforms the skeletal element that is subsequently replaced with bone (Cancedda et al., 1995; DeLise et al., 2000). Chondrocytes move through a progressive series of differentiation steps establishing resting, proliferating, prehypertrophic, and hypertrophic subpopulations. The regulation of the sequential transition from stages of proliferation to hypertrophic differentiation and the formation of bone are dependent on an array of molecular mechanisms acting in a distinct spatial and temporal manner (Erlebacher et al., 1995).

Of the molecular mechanisms regulating endochondral bone formation, TGF- β signaling is thought to play an important role in the regulation of chondrocyte proliferation and hypertrophic differentiation (Serra et al., 1997, 1999). TGF- β receptors have been detected in both chondrocytes and perichondrium, the mesenchymal cells that surround the cartilage element (Sakou et al., 1999; Serra et al., 1999). Targeted, germ-line deletion of the *Tgfb2* gene results in several skeletal abnormalities including shortened long bones, bifurcation of the sternum, as well as defects in the closure of the neural arches in the lumbar regions of the spine (Sanford et al., 1997). *Tgfb1* and *Tgfb3*-null mice do not present with similar skeletal defects (Kaartinen et al., 1995; Kulkarni et al., 1993; Proetzel et al., 1995; Shull et al., 1992). The mechanistic basis by which targeted deletion of the *Tgfb2* gene results in these phenotypes is not known,

but the data suggest an important role for the TGF- β 2 ligand in vertebral and long bone development.

In addition to skeletal defects, *Tgfb2*-null mice exhibited alterations in the development of almost every organ system (Sanford et al., 1997). The mice die at birth primarily due to failure of the cardiovascular system. Furthermore, TGF- β 1 and TGF- β 3 are also present in the skeletal system, and each isoform may compensate for loss of another making it difficult to access the role of TGF- β signaling in all aspects of skeletal development. Mice that carry null mutations in the receptors do not survive past embryonic day 10.5 (Oshima et al., 1996), again making it difficult to study the role of TGF- β signaling in skeletal development. The ability to specifically study the role of TGF- β signaling in the developing skeleton has recently been made possible using the Cre/loxP recombinase system (Sauer, 1998). In this study, we characterize the development of the skeleton in mice expressing Cre under the control of the *Col2a* promoter (*Col2acre*) (Ovchinnikov et al., 2000) crossed to mice in which exon 2 of the *Tgfb2* gene was flanked with bacterial loxP sites (Chytil et al., 2002). In addition to the tissue specificity that this system affords, it has other significant advantages. TGF- β signaling by all three isoforms is effectively blocked, relieving any potential compensatory effect by the other isoforms that may exist in a knockout study of a single isoform. The results presented here indicate that (1) *Tgfb2* is required for normal development of the axial skeleton, but it does not act at the level of chondrocyte differentiation; (2) *Tgfb2* is required for normal development of the IVD and most likely plays a role in defining boundaries between the vertebral body and IVD, and (3) *Tgfb2* is not required in *Col2a* expressing cells for normal embryonic development of the long bones; however, it may play a role in regulating postnatal long bone development.

Materials and methods

Identification of transgenic mice

Cre is the 38-kDa product of the *cre* (cyclization recombination) gene of bacteriophage P1 and is a site-specific DNA recombinase of the INT family. Cre recognizes a 34-bp site on the P1 genome called loxP (locus of X-over of P1) and efficiently catalyzes reciprocal conservative DNA recombination between pairs of loxP sites. Mice generated with a loxP-flanked gene may be mated to transgenic mice with Cre under the control of a promoter with a desired temporal or tissue specificity. This mating will yield a double transgenic mouse where the loxP-modified gene has been deleted in those tissues in which the Cre transgene has been expressed.

For timed pregnancies, specific matings were set up in the afternoon, and the mice were checked for vaginal plugs the next morning. Noon on the day of the vaginal plug was

considered 0.5 days of gestation. The genotype of transgenic mice was determined by PCR analyses of genomic DNA isolated from mouse tails by proteinase K digestion and phenol/chloroform extraction. Col2a-Cre mice were obtained from Jackson Laboratories (Bar Harbor, Maine; (Ovchinnikov et al., 2000)). PCR for the Cre transgene was performed using the following primers: Cre5' (TGC TCT GTC CGT TTG CCG) and Cre3' (ACT GTG TCC AGA CCA GGC). Genomic DNA was amplified for 34 cycles of denaturation at 94°C for 30 s, annealing at 58°C for 60 s, and elongation for 90 s at 72°C in reaction buffer containing 2.5 mM MgCl₂, 1 × PCR buffer (HotMaster Taq buffer, Eppendorf), 0.2 mM dNTPs (Pharmacia, Uppsala, Sweden), 0.2 μM each primer. *Tgfb2-loxP* mice were obtained from Dr. H.L. Moses, Vanderbilt University, Nashville, TN (Chytil et al., 2002). The loxP allele was identified using the following primers: lox1F (TAA ACA AGG TCC GGA GCC CA) and lox1R (ACT TCT GCA AGA GGT CCC CT). Two bands can be detected with these primers. One band represents the wild-type allele (420 bp), and another represents the loxP allele (540 bp). The presence of both bands indicates a mouse that is heterozygous for the loxP allele. To detect recombination and the subsequent loss of the floxed allele (*Tgfb2* exon2), the lox1F and lox2R (AGA GTG AAG CCG TGG TAG GT) primers were used. Successful recombination was confirmed by the presence of a 610-bp band.

Whole mount skeletal preparation

Skeletons from 13.5-, 15.5-, and 17.5-day p.c. fetal mice were double-stained for cartilage and bone with Alcian blue and alizarin red (Kimmel and Trammell, 1981; McLeod, 1980). All carcasses were skinned and fixed in 95% ethanol for 72 h. Samples were then placed in 95% ethanol-Alcian blue solution (20 h) for cartilage staining, followed by a 95% ethanol wash (8 h), and maceration in 0.35% KOH overnight. Bone was stained with alizarin red S in 0.2% KOH (4 to 6 h) followed by clearing in 95% ethanol: glycerin (1:1). Whole mount skeletal preparations of postnatal mice were prepared according to Selby (1987). Briefly, mice were skinned and eviscerated then soaked in a series of KOH solutions over several days. Skeletal tissue was stained with 0.004% alizarin red in 1.9% KOH. The specimens were cleared in glycerin, benzyl alcohol, and ethanol and then stored in glycerin. Images were taken using an Olympus SZX12 dissecting microscope and Magnafire digital camera.

Histology and in situ hybridization

Mouse embryos were fixed overnight at 4°C in 4% paraformaldehyde. Specimens were then dehydrated and embedded in paraffin. Sections were cut at a thickness of 5 μm and mounted on Superfrost Plus slides (Fisher). For routine histological analysis, sections were stained with

hematoxylin and eosin using standard procedures. Images of the sections were taken using an Olympus BX-51 upright microscope and Macrofire digital camera.

In situ hybridization was performed as described (Pelton et al., 1990) on sections from mouse embryos. Embryos were fixed overnight at 4°C in fresh 4% paraformaldehyde then dehydrated through ethanol and xylene then embedded in paraffin. Sections were hybridized to ³⁵S-labeled sense and antisense riboprobes. The Cre probe was made from the 720-bp PCR fragment generated for the Cre transgene. The fragment was subcloned into the pGEM-T Easy (Promega) cloning vector. The type IIAB (pEL111) and type IIB (pSOK) collagen probes were kindly provided by Dr. K.S.E. Cheah (Cheah et al., 1991; Ng et al., 1993). The Col10a probe (a gift from Dr. Bjorn Olsen, Harvard Medical School, Boston, MA) was a 650-bp *Hind*III fragment containing 360 bp of noncollagenous (NC1) domain and 260 bp of 3'-untranslated sequence of the mouse *Col10a1* gene in pBluescript (Apte et al., 1992). Slides were exposed to photographic emulsion at 4°C for 1 week (Cre) or 3 days (Col10a, Col2a) then developed with D19 developer, fixed in 1% acetic acid and cleared in 30% sodium thiosulfate. Sections were counterstained with 0.2% toluidine blue. Images of the sections were obtained using an Olympus BX51 microscope and Macrofire digital camera. Images were taken under bright field and dark field illumination.

Bromo Deoxyuridine (BrdU) labeling

Pregnant dams were injected with 0.06 g BrdU/g mouse. The females were sacrificed 2.5 h after injection, and the embryos were genotyped, fixed, embedded, and sectioned. Sections were deparaffinized, denatured in 2 N HCl for 20 min at 37°C, and neutralized in 1% boric acid/0.0285% sodium borate, pH 7.6. Next, the sections were treated with 0.005 mg trypsin/ml 0.05 M Tris, pH 7.6, for 3 min at 37°C and washed three times in PBS. Immunostaining was then performed using Vectastain Elite staining kit (Vector Laboratories) as described by the manufacturer. A rat mAb directed to BrdU (Harlan) was used as the primary antibody at a 1:200 dilution. Cy3-conjugated avidin (Vector Laboratories) was substituted for the avidin–biotin–peroxidase complex in the Vector Elite kit. Excess Cy3-conjugated avidin was removed from the sections by washing three times for 10 min each in PBS at room temperature. Sections were then counterstained with YoPro (Molecular Probes), a green nuclear stain. The sections were immediately mounted with Aquapoly mount (Poly Sciences). Fluorescence was observed and imaged using an Olympus BX-51 upright microscope with a Macrofire digital camera. Labeled and unlabelled cells were counted, and the percent labeled nuclei were calculated for each group. The mean and standard deviation were calculated, and significant differences between each group were calculated using Student *t* test. Results that were statistically different were determined by a *P* value of <0.05.

Results

Generation of mice with a conditional deletion of *Tgfr2*

To address the question of the role of TGF- β signaling in skeletal development, we generated mice with targeted deletion of *Tgfr2* specifically in *Col2a* expressing cells using the Cre/lox recombinase system. T β RII mediates signaling by all three isoforms of TGF- β (Derynck, 1994; Shi and Massague, 2003; Ten Dijke et al., 1996). We chose to delete *Tgfr2* rather than individual ligands to avoid potential problems with functional redundancy. Furthermore, since the ligands are secreted proteins, they are not expected to act in a cell autonomous manner, whereas targeted deletion of the receptor should result in altered signaling in only those cells in which the receptor has been deleted. In addition, germ line deletion of *Tgfr2* results in early embryonic lethality (Oshima et al., 1996). Disruptions to skeletal development are not expected to be lethal until very late in gestation or after birth. The expression pattern of Cre in *Col2a* mice was shown previously using the ROSA-26-Cre reporter mouse strain in which Cre activity results in expression of a β -galactosidase reporter (Ovchinnikov et al., 2000). In somites, Cre activity was first detected at 9.5 days of gestation in the sclerotome, the population of cells that will become the vertebrae. Cre expression continued at 12.5 days in the vertebral anlagen. Low levels of expression were also detected in the notochord, an important signaling center in the development of the ventral portion of the vertebrae. Cre was also detected in all cartilage of the long bones but was not detected in the perichondrium (Long et al., 2001; Ovchinnikov et al., 2000). Mice expressing Cre under the control of the *Col2a* promoter (*Col2a*Cre; (Ovchinnikov et al., 2000) were crossed to mice in which the *Tgfr2* gene was flanked with bacterial loxP sites (Chytil et al., 2002). This allele generates a null mutation by deletion of exon 2. Mice homozygous for the loxP allele (*Tgfr2*^{loxP/loxP}) were crossed to mice that were heterozygous for the loxP allele and hemizygous for the *Col2a*-Cre transgene (*Col2a*Cre^{+/-}; *Tgfr2*^{loxP/+}). The mice derived from these crosses were the basis of all subsequent studies. Mice with the genotype *Col2a*Cre^{+/-}; *Tgfr2*^{loxP/loxP} were the experimental group, whereas mice negative for Cre (*Col2a*Cre^{-/-}; *Tgfr2*^{loxP/+} or *Col2a*Cre^{-/-}; *Tgfr2*^{loxP/loxP}) were used as controls. Haploinsufficiency effects were not detected in *Col2a*Cre⁺; *Tgfr2*^{loxP/+} mice, which were the same as Cre controls in the following experiments.

Expression of Cre mRNA in was confirmed in 15.5-day embryos by radioactive in situ hybridization (Fig. 1A). The expression pattern conformed to what was previously observed (Ovchinnikov et al., 2000). Radioactive in situ hybridization can be used to determine expression of mRNA in individual cells. At high magnifications, hybridization to Cre mRNA was clearly seen under bright field illumination in all chondrocytes (Fig. 1A). Importantly for this study, Cre mRNA was not detected in the perichondrium around the long bones (Fig. 1A). To

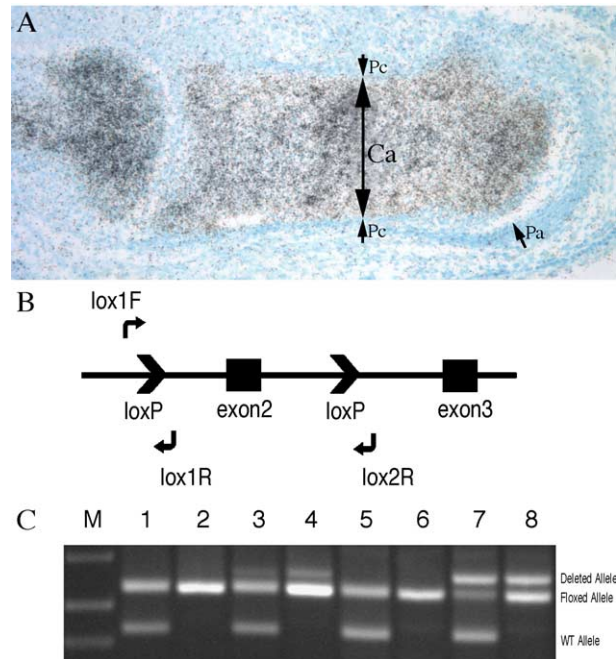


Fig. 1. Cre expression and *Tgfr2* exon 2 deletion. (A) Cre mRNA was detected by in situ hybridization in sections from Cre-positive embryos at 15.5 days of gestation. Cre expression was detected as black grains in bright field images of sections hybridized with an antisense riboprobe directed to the Cre transgene. No background hybridization was detected with a Cre sense riboprobe or on sections from Cre-negative mice (not shown). Cre mRNA was detected in cartilage (Ca) of developing endochondral bones, but Cre mRNA was not detected in the perichondrium (Pc) or in the periarticular perichondrium (Pa). A section through a phalanx is shown. The slides were counterstained with toluidine blue. (B) A map of the loxP sites flanking *Tgfr2* exon 2 is shown. The localization of the primers used in this study is indicated. (C) PCR detection of the deleted allele. DNA isolated from the tip of adult mouse tails (1–4) or from the tibia and fibula of E15.5-day mouse embryos (5–8) was used to detect deleted exon 2 (primers lox1F and lox2R), the floxed allele, and the wild-type allele (primers lox1F and lox1R) in the presence (3, 4, 7, 8) and absence (1, 2, 5, 6) of Cre. The 610-bp band representing the exon 2 deletion is only seen in Cre-positive samples.

confirm the deletion of *Tgfr2* and assess the level of Cre directed recombination in cartilage, a PCR based assay was used (Figs. 1B, C). DNA was isolated from tibia and fibula of Cre- and Cre⁺ 15.5-day embryos. DNA was used in a PCR assay that allowed for the simultaneous detection of the wild-type (420 bp), loxP (540 bp), and deleted (610 bp) *Tgfr2* alleles. Specific primers for the Cre transgene were used to determine the presence of Cre. In the presence of Cre, a high degree of recombination was detected in mice heterozygous and homozygous for the floxed *Tgfr2* allele. The continued presence of undeleted allele may be attributed to the heterogenous pool (total bone) of cells used for the extraction of the DNA. Nevertheless, the results indicate that there is a significant level of recombination in the skeletal elements.

To determine if mice with the tissue-specific *Tgfr2* deletion survive, the genotypes of newborn offspring as

well as embryos at 17.5 days of gestation were determined so that the actual versus the expected Mendelian ratio of each genotype could be calculated. The $\text{Col2acre}^{-/-}; \text{Tgfb}r2^{\text{loxP/loxP}}$ X $\text{Col2acre}^{+/-}; \text{Tgfb}r2^{\text{loxP/loxP}}$ crosses should yield a 1:1:1:1 ratio for each of the possible genotypes: $\text{Col2acre}^{-/-}; \text{Tgfb}r2^{\text{loxP/loxP}}$; $\text{Col2acre}^{-/-}; \text{Tgfb}r2^{\text{loxP/loxP}}$; $\text{Col2acre}^{+/-}; \text{Tgfb}r2^{\text{loxP/loxP}}$; $\text{Col2acre}^{+/-}; \text{Tgfb}r2^{\text{loxP/loxP}}$. The ratio of newborn mice for the experimental group was far below expected (7% versus 25%) suggesting that many of the mice lacking *Tgfb*2 did not survive postnatally. In contrast, the ratio of $\text{Col2acre}^{+/-}; \text{Tgfb}r2^{\text{loxP/loxP}}$ embryos at 17.5 days of gestation was as expected, suggesting that mice in which *Tgfb*2 had been deleted in *Col2a* expressing cells survived until late in gestation.

Cranial and axial skeleton

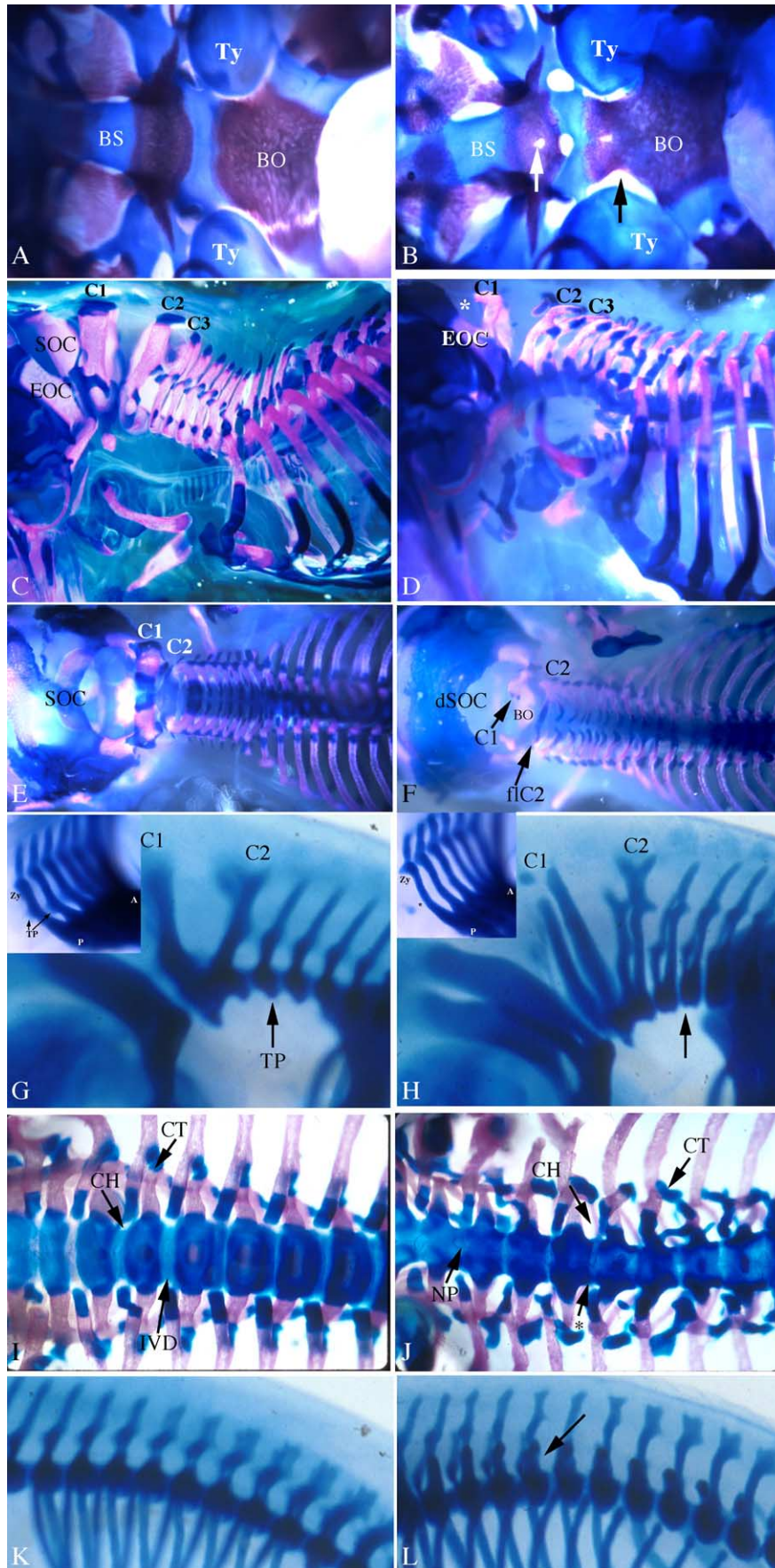
Skeletons from 7 to 10 embryos of each Cre/Lox genotype at 17.5, 15.5, and 13.5 days of gestation were stained with alizarin red and Alcian blue to define the mineralized areas and the cartilage, respectively (Figs. 2, 3 and 8). Skeletons were also prepared from *Tgfb*2-null mice at E17.5 days so that the phenotypes resulting from the loss of the one ligand (TGF- β 2) in every cell type could be compared to that in mice in which the receptor is deleted in a subset of cells.

Many of the bones in the skull develop through the process of intramembranous ossification, which does not involve a cartilage template. In contrast, bones at the base of the skull including the occipital and basisphenoid bones develop through endochondral ossification. As expected, craniofacial bones that develop through intramembranous ossification were similar in Cre- and $\text{Col2acre}^{+/-}; \text{Tgfb}r2^{\text{loxP/loxP}}$ mice. The only malformations detected in the skull were in the basioccipital, exoccipital, supraoccipital, and basisphenoid bones (Figs. 2A–F). The basioccipital, exoccipital, and basisphenoid bones were reduced in overall size. The supraoccipital bone was missing. Large spaces between the bones at the base of the skull were observed. In addition, large holes were present in the basioccipital and basisphenoid bones. The craniopharyngeal canal was greatly enlarged in the $\text{Col2acre}^{+/-}; \text{Tgfb}r2^{\text{loxP/loxP}}$ mice relative to

the Cre-negative controls in which it is not detectable at this stage. In contrast to what is observed in *Tgfb*2-null mice (Sanford et al., 1997), development of facial bones and the palate, which is derived from neural crest, was not altered in $\text{Col2acre}^{+/-}; \text{Tgfb}r2^{\text{loxP/loxP}}$ mice.

Many defects were observed in the formation of the vertebrae (Figs. 2 and 3). All of the vertebrae were reduced in size. That is, each vertebra was thinner (anterior to posterior) than controls (Figs. 2C, D and 3C, D). The vertebral bodies were also reduced in size and irregularly shaped (Figs. 2I, J and 3G, H). Vertebrae were irregularly spaced, most likely a consequence of reduced or absent intervertebral discs (Figs. 2I, J and 3I–L). C1 and C2 were severely affected (Figs. 2C–F). The thicknesses of C1 and C2 were reduced to the size of the other cervical vertebrae. In some $\text{Col2acre}^{+/-}; \text{Tgfb}r2^{\text{loxP/loxP}}$ mice, the dorsal edges of C1 remained open, or the Alcian blue-stained dorsal section of the neural arch was completely absent, whereas at this stage in control mice, the dorsal edges of C1 had fused to complete the neural arch (Figs. 2C–F). The neural arch of C2 in experimental mice was split transversely in two, so that when viewed dorsally, it appeared as if there was an extra vertebra or that there was a “floating” neural arch without a vertebral body (Figs. 2E, F). All of the dorsal edges of the cervical vertebrae were irregularly shaped and angled anteriorly so that they did not come together in the midline as seen in the controls (Figs. 2E–F). Transverse processes were reduced or absent, and the anterior zygapophyses were missing, while the posterior zygapophyses were irregularly shaped in experimental animals (Figs. 2C, D). Alterations in the size and shape of the cervical vertebrae relative to controls could be seen in skeletal preparations from $\text{Col2acre}^{+/-}; \text{Tgfb}r2^{\text{loxP/loxP}}$ embryos as early as 13.5 days of gestation (Figs. 2G, H). Dorsal edges of the thoracic vertebrae were also irregularly shaped compared to controls so that they did not come together in the midline. Anterior zygapophyses were reduced or absent. In addition, alterations in articulation of the costal head and costal tubercle with the vertebral body and transverse process of the thoracic vertebrae were detected (Figs. 2I, J). Alterations in the shape of the thoracic vertebrae $\text{Col2acre}^{+/-}; \text{Tgfb}r2^{\text{loxP/loxP}}$ mice could be detected

Fig. 2. Skeletal preparations, anterior. Alizarin red-/Alcian blue-stained skeletons were prepared from control $\text{Col2acre}^{-/-}; \text{Tgfb}r2^{\text{loxP/+}}$ (A, C, E, G, I, K) and experimental $\text{Col2acre}^{+/-}; \text{Tgfb}r2^{\text{loxP/loxP}}$ (B, D, F, H, J, L) embryos at 13.5 (G, H, K, L) or 17.5 (A–F, I, J) days of gestation. Dorsal view through the skull (A, B). Defects in the basioccipital (BO) and basisphenoid (BS) bones were detected in experimental mice. Large spaces are seen between the BO/BS bones and the tympanic bones (Ty). Holes were detected in the BO and BS bones (examples with white arrows). Sagittal view of the cervical region (C, D). Alterations were also detected in the exoccipital bone (EOC) and in the cervical vertebrae (C1, C2). The supraoccipital bone (SOC) was missing in experimental mice (white asterisk). Dorsal view of cervical region (E, F). The supraoccipital bone was missing in experimental mice (dSOC). The Alcian blue-stained dorsal portion of C1 was missing. The dorsalmost edge of C1 is demarcated with an arrow (C1). The basioccipital bone (BO) can be seen underneath. The dorsal edges of the other neural arches did not come together in the midline, and the neural arch of C2 was split giving rise to a “floating” arch (fC2). Sagittal view of the cervical region at E13.5 days (G, H). Defects in the cervical vertebrae could be seen as early as E13.5 days. Transverse processes were also missing in the cervical region (arrow, TP). Insets show a higher magnification view of the transverse process (TP) in wild-type mice and the area of the missing TP in experimental mice (asterisk). Zygapophyses are also visible (Zy). Ventral view of the thoracic region (I, J). Defects were detected in the articulation of the costal head (CH) and costal tubercle (CT) with thoracic vertebrae. In controls, the IVD are clearly demarcated. In the experimental animals, the nucleus pulposus (NP) is visible, but the light blue staining of the annulus fibrosus is not seen (asterisk). See also Figs. 4G and H regarding the IVD. Sagittal view of the thoracic region in E13.5-day embryos (K, L). Alterations in the transverse processes and the articulation of the ribs can be seen as early as 13.5 days of gestation (arrow).



as early as 13.5 days of gestation (Figs. 2K, L). The lumbar vertebrae demonstrated similar alterations (Fig. 3). The dorsal edges of the neural arches did not come together at the midline (Figs. 3A, B). Some of the dorsal edges of the lumbar neural arch were notched, and others had holes in them. The anterior zygapophyses were also missing (Figs. 3C, D). Kinks were observed at the lumbar region in the vertebral columns of several $Col2acre^{+/-};Tgfb2^{loxP/loxP}$ mice. The kinks are most likely due to variations in the size and shape of vertebral bodies as well as the lack of intervertebral discs. The phenotype of $Col2acre^{+/-};Tgfb2^{loxP/loxP}$ mice in the lumbar vertebrae resembled that seen in $Tgfb2$ -null mice with regards to the failure of the dorsal neural arch to fuse and lack of zygapophyses (Figs. 3E, F and Sanford et al., 1997). In contrast, intervertebral discs were intact in $Tgfb2$ -null mice, and alterations were not detected in the formation of cervical or thoracic vertebrae (Sanford et al., 1997 and data not shown). The results indicate that $Tgfb2$ is required for proper formation of the base of the skull and vertebrae.

Next, we examined the histology of the cartilage in the cranial and axial skeleton to determine if there were any alterations in the organization of the cartilage (Fig. 4). Transverse and sagittal sections from embryos at 15.5 and 17.5 days of gestation were stained with hematoxylin and eosin. Transverse sections from the same level anterior to posterior or sagittal sections from similar levels lateral to medial were compared. No differences were detected in the organization of the cartilage in control versus $Col2acre^{+/-};Tgfb2^{loxP/loxP}$ embryos in the basioccipital or basisphenoid bones or in the vertebrae. Furthermore, significant alterations were not detected in the extent of endochondral bone formation as measured by the expression patterns of $Col2a$ and $Col10a$ mRNA in each skeletal element (Fig. 5 and data not shown) consistent with the comparable level of alizarin red staining observed in the whole mount skeletal preparations. In the vertebrae, the extent of endochondral bone formation can best be seen as the expression boundary for $Col10a$ mRNA measured by radioactive in situ hybridization in the lateral ossification center (Figs. 5C, D, arrows). The $Col10a$ expression boundary in the central ossification center of the vertebral body varies significantly with slight variations in anterior to posterior position and is therefore not a good indication

of the extent of endochondral differentiation. The results suggest that, although $Tgfb2$ is required for proper formation of the base of the skull and the axial skeleton, the program of endochondral ossification is not affected.

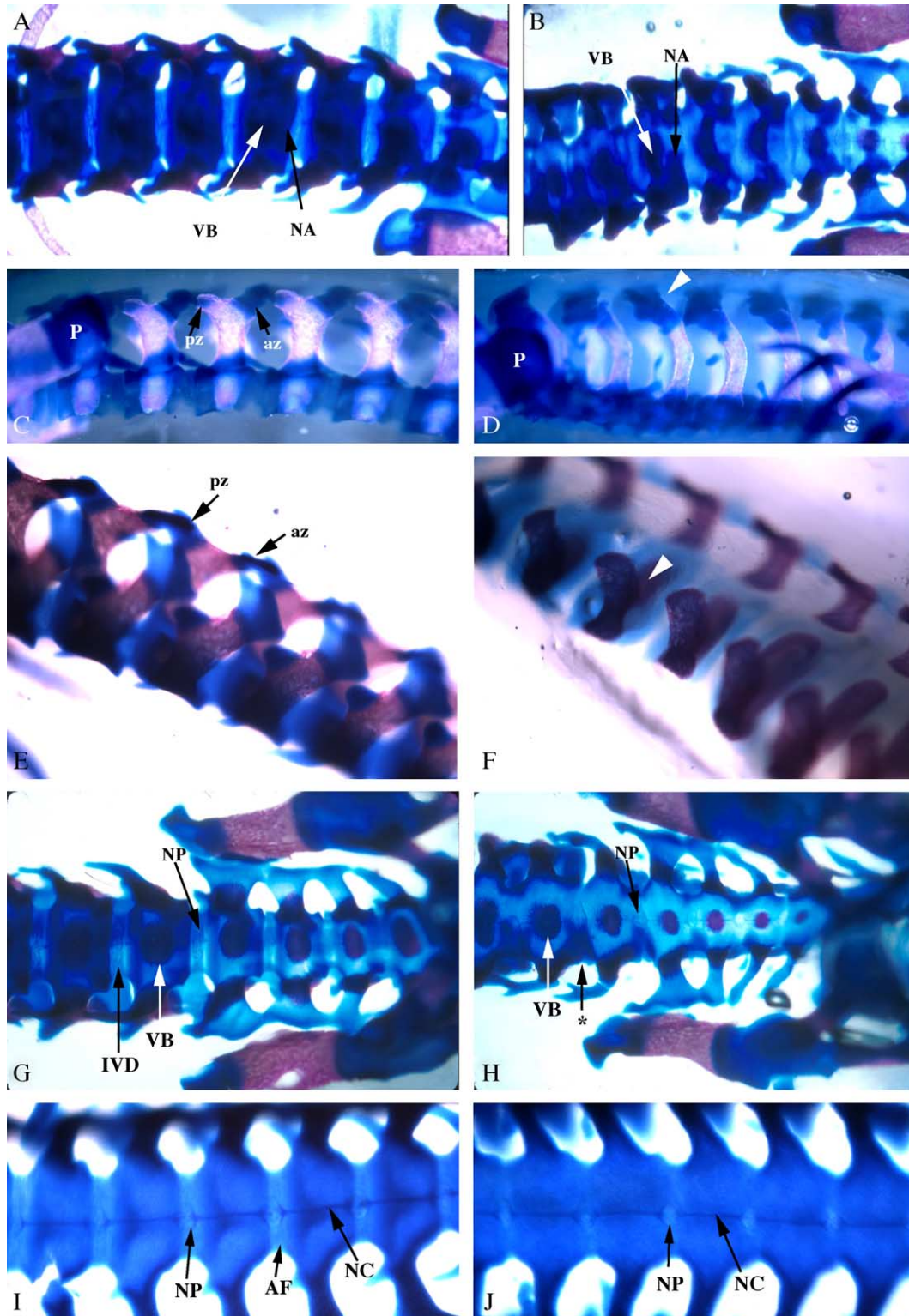
Intervertebral discs

The intervertebral discs are derived from the ventral part of the sclerotome (Paavola et al., 1980; Rufai et al., 1995; Theiler, 1988). Initially, the sclerotome surrounds the notochord forming a continuous unsegmented tube. At approximately 12 days of gestation in the mouse, the tube is segmented into regions of condensed and loose mesenchyme. The condensed mesenchyme will form the intervertebral disc (IVD), and the loose mesenchyme will undergo chondrogenesis to form the vertebral body. Notochord cells are then removed from the vertebral region and expand into the IVD region to form the nucleus pulposus. The region around the nucleus pulposus forms the annulus fibrosus, which provides the structural properties of the IVD. The annulus fibrosus is divided into an inner portion containing hyaline cartilage and an outer portion that is more fibrous. In skeletal preparations of normal E15.5- and E17.5-day mice, the nucleus pulposus can be seen outlined in the middle of the IVD region (Figs. 2I and 3G, I). The IVD region appears light blue in between the dark blue of the vertebral bodies. By E17 days, a light blue area of IVD is clearly demarcated, and red areas of mineralization in the vertebral body are observed. In $Col2acre^{+/-};Tgfb2^{loxP/loxP}$ mice, the IVD region does not develop properly (Figs. 2I, J and 3G–J). The nuclei pulposi form and can be seen at E15.5 days; however, the light blue staining of the annuli fibrosus is not detectable (Fig. 3J). At E17.5 days, notochord cells have been removed from the vertebral bodies, and the nuclei pulposi are present in the IVD region but have not expanded to the same extent as observed in controls (Figs. 2J and 3H). The light blue area of the annuli fibrosus is not observed, and dark blue and red staining indicative of mineralization is seen around the nuclei pulposi (Fig. 3H). Alterations in the inner annulus were detected in hematoxylin and eosin-stained sagittal sections from E15.5-day $Col2acre^{+/-};Tgfb2^{loxP/loxP}$ mice when compared to sections from control littermates (Figs. 6A–D). In the more

Fig. 3. Skeletal preparations, posterior. Alizarin red-/Alcian blue-stained skeletons were prepared from control $Col2acre^{-/-};Tgfb2^{loxP/+}$ (A, C, G, I), experimental $Col2acre^{+/-};Tgfb2^{loxP/loxP}$ (B, D, H, J), $Tgfb2^{+/+}$ (E), and $Tgfb2^{-/-}$ (F) embryos at 15.5 (I, J) or 17.5 (A–H) days of gestation. Dorsal view of the lumbar region (A,B) shows the staining of the ossification center of the vertebral body (VB, white arrow) only faintly through the dark blue stain of the neural arch (NA, black arrow) which is completely fused on the dorsal midline in control mice. In the $Col2aCre^{+/-};Tgfb2^{loxP/loxP}$ mouse, the neural arch is not completely formed (NA, black arrow), and the ossification center of the vertebral body below (VB, white arrow) is seen unobstructed. Sagittal view of the lumbar region (C–F). The pelvis bone (P) is shown on the left. The lateral part of the neural arch can be seen in control mice (C, E) with anterior and posterior zygapophyses (az, pz). In $Col2acre^{+/-};Tgfb2^{loxP/loxP}$ (D) and $Tgfb2$ -null mice (F), the zygapophyses are reduced or missing (white arrow heads). Ventral view of the lumbar/sacral region (G, H). The IVD and vertebral bodies (VB) are clearly demarcated in the ventral view of the lumbar/sacral region in the control mouse. The nucleus pulposus has expanded across the region of the IVD (NP). IVD are missing or incomplete in experimental mice. The nucleus pulposus (NP) is present, but it has not expanded. The light blue staining of the annulus fibrosus is not detected, and the area of the IVD appears mineralized (*). Ventral view of the lumbar region in E 15.5-day mice (I, J). The nucleus pulposus (NP) is visible in both control and experimental IVD. The annulus fibrosus (AF) is not visible in the experimental mice.

lateral sections of control mice (Fig. 6A), the area of the IVD was larger than that of the vertebral body, and the concentric arrangement of cells of the inner annulus was clearly demarcated. In more medial sections from control embryos (Fig. 6C), the nuclei pulposi were seen along with a clear demarcation between the condensed cells of the IVD

region and the area of chondrogenesis in the vertebral body. In the lateral sections from *Col2acre^{+/-};Tgfb β 2^{loxP/loxP}* mice (Fig. 6B), the areas of IVD and vertebral body were not clearly demarcated except that the outer annulus was visible at the region of the IVD. In the medial sections, the nuclei pulposi and outer annuli were visible, but the



cells of the inner annuli were not clearly demarcated (Fig. 6D). Disorganization in the inner annulus was also observed at 17.5 days of gestation (Figs. 6E, F). The results suggest that T β RII is required for proper development of the IVD.

To determine if there were alterations in the organization or differentiation of the IVD, the expression domain of the mRNAs for two alternatively spliced forms of type II collagen was determined by in situ hybridization of sagittal sections from 15.5-day embryos (Fig. 7). Type II collagen is expressed as two forms by differential splicing of the initial transcript (Ng et al., 1993; Sandell et al., 1991). The two forms are distinguished by the presence or absence of exon 2. Collagen type IIA includes sequences from exon 2, while type IIB does not. Type IIA is expressed in chondroprogenitor cells and some noncartilagenous tissues including the notochord and intervertebral area (Ng et al., 1993; Sandell et al., 1991). Type IIB is made by chondrocytes. One probe used for in situ hybridization contains exon 1 of *Col2a1* and therefore recognizes mRNA for both types IIA and IIB collagens (Cheah et al., 1991). The other probe contains exon 2 and only recognizes mRNA for type IIA collagen (Ng et al., 1993). Segmented areas of types IIB and IIA expression were clearly demarcated in sections from *Col2acre^{-/-};Tgfb2^{loxP/loxP}* mice (Figs. 7A, C). High levels of hybridization were seen in the future vertebral body in control mice using the type IIB probe. Low levels were detected in the region of the future IVD. Expression of type IIA mRNA was seen clearly in the future IVD area of control mice with the type IIA specific probe. In contrast, the segmentation of types IIB and IIA expression was not clearly demarcated in sections from *Col2acre^{+/-};Tgfb2^{loxP/loxP}* embryos (Figs. 7B, D). Hybridization to both probes was diffuse throughout the axial region. The results suggest that *Tgfb2* has a role in defining the boundaries between the vertebral and IVD regions.

Appendicular skeleton

Next, the development of the appendicular skeleton was studied. The total lengths of long bones from 7 to 10 skeletons prepared from E17.5-day mice were measured, and the averages were calculated (Fig. 8A and Table 1). The amount (percent) of mineralization was calculated as the length of the alizarin red-stained portion of the bone over the total length of the bone. Significance was

determined using a Student *t* test. Significant differences between control and experimental groups in the percent of mineralization were not detected. Only marginal differences were detected in the total length of the long bones. Proliferation in chondrocytes as measured by the incorporation of BrdU was not significantly different in control and *Col2acre^{+/-};Tgfb2^{loxP/loxP}* mice, although there seemed to be a trend towards increased proliferation (Table 2). Furthermore, alterations were not detected by standard histology in the organization of the growth plate (Figs. 9A, B) nor were alterations in the expression domain of *Col10a* detected by in situ hybridization (Figs. 9C, D). In contrast, significant differences in both the length and mineralized area of long bones from *Tgfb2*-null mice were noted (Fig. 8B and Table 1). The mineralized area was increased in all bones examined. Significant decreases in total bone length were identified in the more distal bones, tibiae and radii. Incorporation of BrdU into radii and metatarsals was significantly decreased in *Tgfb2*-null mice when compared to wild-type controls (Table 2). An additional contrast to the *Tgfb2*-null mice is that the deltoid tuberosity was intact in *Col2acre^{+/-};Tgfb2^{loxP/loxP}* mice (Figs. 8A, B). The data suggest that any role of *Tgfb2* in embryonic long bones is mediated indirectly through cells that do not express *Col2a-Cre*, likely cells in the perichondrium as previously suggested (Alvarez et al., 2001, 2002).

It should be noted that the length of long bones in the few *Col2acre^{+/-};Tgfb2^{loxP/loxP}* mice that survived birth was progressively decreased relative to control littermates (Figs. 8C, D and Table 3). Alterations were not detected at postnatal day 3; however, alterations were detected starting at postnatal day 7 to 12 weeks. The length of proximal long bones in the arms and legs was reduced, while the length of metatarsal bones and the length of the skull were not altered. This suggests that the reduced size of the proximal long bones is not due to isometric or systemic effects on growth presenting the possibility that the role of T β RII varies in embryonic and postnatal development. Differences were not observed in the percentage of the mineralized area in the long bones of *Col2acre^{+/-};Tgfb2^{loxP/loxP}* mice at any age.

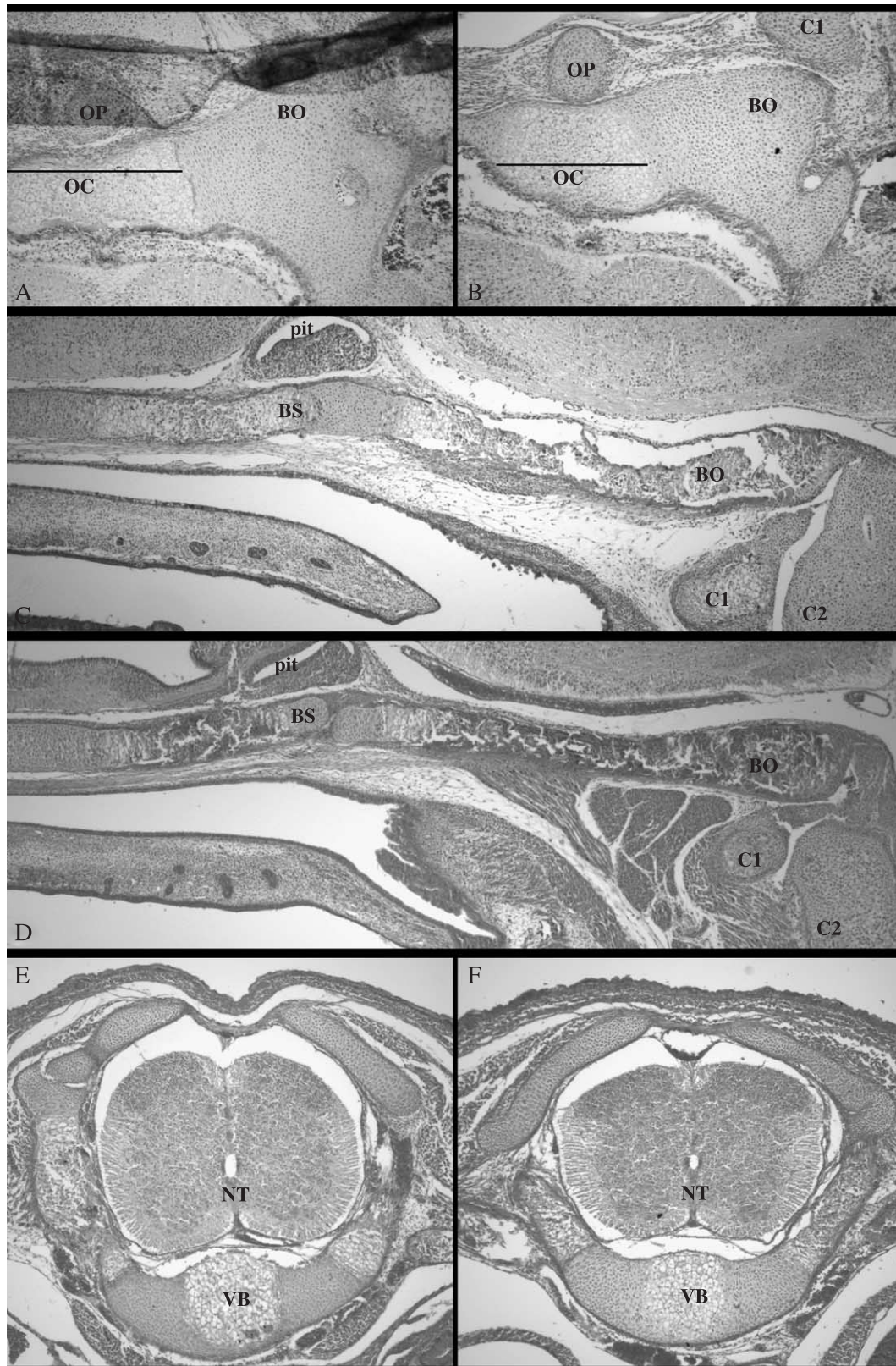
Discussion

The results presented here provide information regarding mechanisms of development of the axial and appendicular

Fig. 4. Cartilage histology, skull and vertebrae. Transverse sections through the basioccipital bone (A, B). Hematoxylin and eosin-stained sections through the basioccipital bone of Cre-negative (A) and *Col2acre^{+/-};Tgfb2^{loxP/loxP}* (B) embryos at 15.5 days of gestation. Cartilage organization in the basioccipital bone (BO) is similar in both conditions. The ossification center (OC) is clearly demarcated, and little to no difference in the extent of hypertrophic differentiation is observed. The odontoid process of C2 (OP) is marked for reference. Sagittal sections through the basioccipital and basisphenoid bones (C, D). Hematoxylin and eosin-stained sections of Cre-negative (C) and *Col2acre^{+/-};Tgfb2^{loxP/loxP}* (D) embryos at 15.5 days of gestation are shown. The basioccipital bone (BO) and basisphenoid bone (BS) show a comparable level of ossification in control and experimental animals. The first and second cervical vertebrae (C1 and C2) and the pituitary gland (pit) are shown for reference. Transverse sections in the mid thoracic vertebrae at the level of the diaphragm, E17.5 days (E, F). The organization of the chondrocytes and the amount of hypertrophic cartilage were not altered in the vertebral bodies (VB) in control (E) versus experimental (F) mice. Neural tube (NT).

skeleton. Specifically, we have demonstrated the necessity of $T\beta RII$ in specific aspects of skeletal development. Mice that express the Cre recombinase under the control of the

Col2a promoter were crossed to mice in which exon 2 of *Tgfb β 2* was flanked with loxP sites, resulting in conditional deletion of *Tgfb β 2*. Skeletal development was compared in



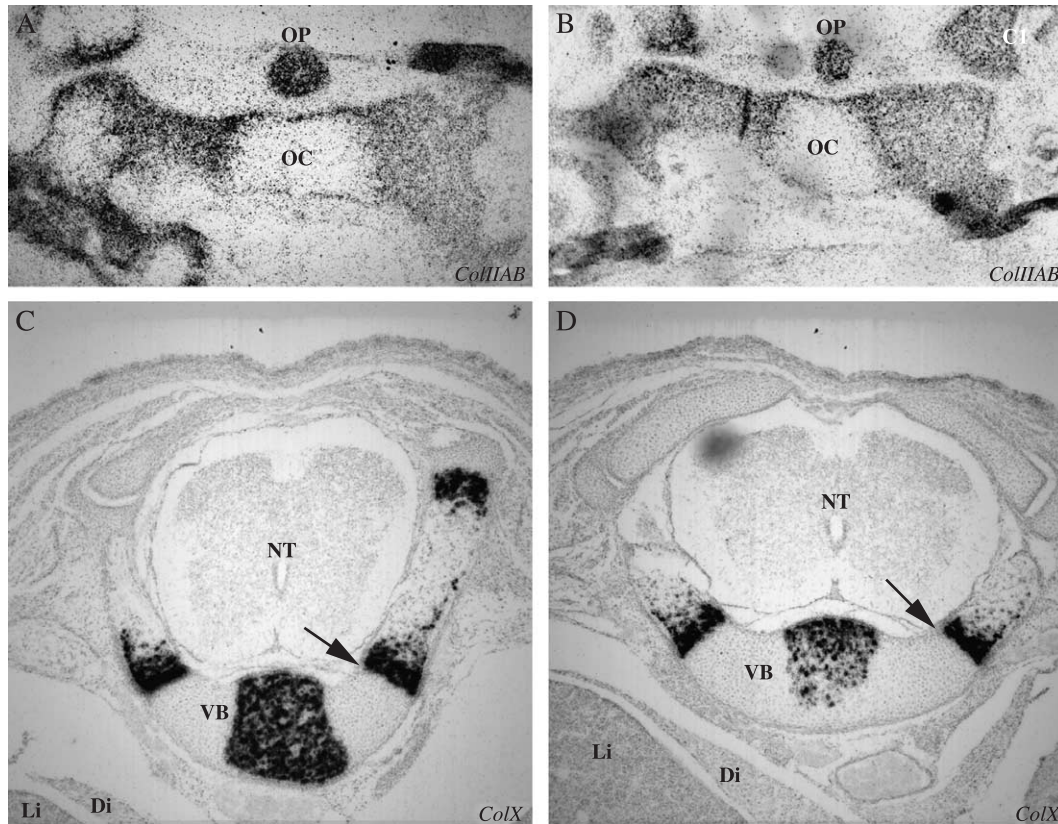


Fig. 5. In situ hybridization for markers of cartilage differentiation, cranial and axial skeleton. Expression of type IIAB collagen in the basioccipital bone (A, B). Transverse sections through the basioccipital bone (BO) from control (A) and experimental mice (B) at 15.5 days of gestation were hybridized to a riboprobe complementary to overlapping sequences from types IIA and IIB forms of collagen (ColIIAB). Hybridization is detected as dark black grains on the bright field image of toluidine blue-stained sections. Similar expression domains were seen under each condition. The odontoid process (OP) is marked for reference. The ossification center (OC) is well demarcated and hybridized strongly to a probe for type X collagen (not shown). Chondrocyte differentiation is comparable in control and experimental mice. Type X collagen expression in the vertebrae (C, D). Sections from vertebrae at the midthoracic level in Cre-negative (C) and Col2acre^{+/-};Tgfb2^{loxP/loxP} (D) embryos at 17.5 days of gestation were hybridized to an antisense riboprobe directed to type X collagen (ColX). The vertebral body (VB) is marked, and the area of ColX expression is seen as dark black grains on the toluidine blue-stained section. The extent of hypertrophic differentiation can be seen as the border of the ColX expression domain on the lateral part of the vertebrae (black arrows). The extent of hypertrophic differentiation is comparable in control and experimental mice. The liver (Li), diaphragm (Di), and neural tube (NT) are marked for reference.

Cre-positive; Tgfb2^{loxP/loxP}, and Cre-negative mice. Multiple defects were found in the development of the base of the skull and the vertebrae. The defects included a reduction in size and alterations in the shape of specific skeletal elements. Lateral and dorsal elements of the vertebrae were missing or irregularly shaped. The size and shape of vertebral bodies were mildly affected, but the intervertebral discs which are also derived from ventral sclerotome were greatly reduced or missing. Surprisingly, defects in the differentiation of chondrocytes were not detected in the axial or appendicular skeleton.

The expected ratio of mice with the conditional deletion survived until late in gestation; however, far fewer mice than expected survived postnatally. The high rate of perinatal lethality in the mice with the conditional deletion is surprising. There are several possibilities for this observation. Some of the defects in the cervical vertebrae and base of the skull are severe, thus the vertebrae may not be able to withstand the stress of the birthing process. Alternatively, the

Cre transgene may be expressed at low levels in embryonic tissues other than cartilage. Although not detected in crosses of Col2acre mice with the ROSA-26 reporter line, low levels of type II collagen have previously been detected in the developing heart (Cheah et al., 1991; Ovchinnikov et al., 2000; Rahkonen et al., 2003; Swiderski et al., 1994). It is possible that a low level of Cre could lead to a sufficient loss of TβRII in the heart to result in lethal defects.

TGF-β in development of the head and spine

The three isoforms of TGF-β are expressed in the developing axial skeleton in distinct and overlapping patterns (Milani et al., 1991; Pelton et al., 1990; Schmid et al., 1991). TGF-β1 mRNA is localized to intersegmental cells at 12.5 days of gestation. By 16.5 days, TGF-β1 mRNA is localized to the ossification centers and perichondrium of vertebrae. TGF-β1 mRNA was not detected in the notochord. At 12.5 days of gestation, TGF-β2 mRNA is

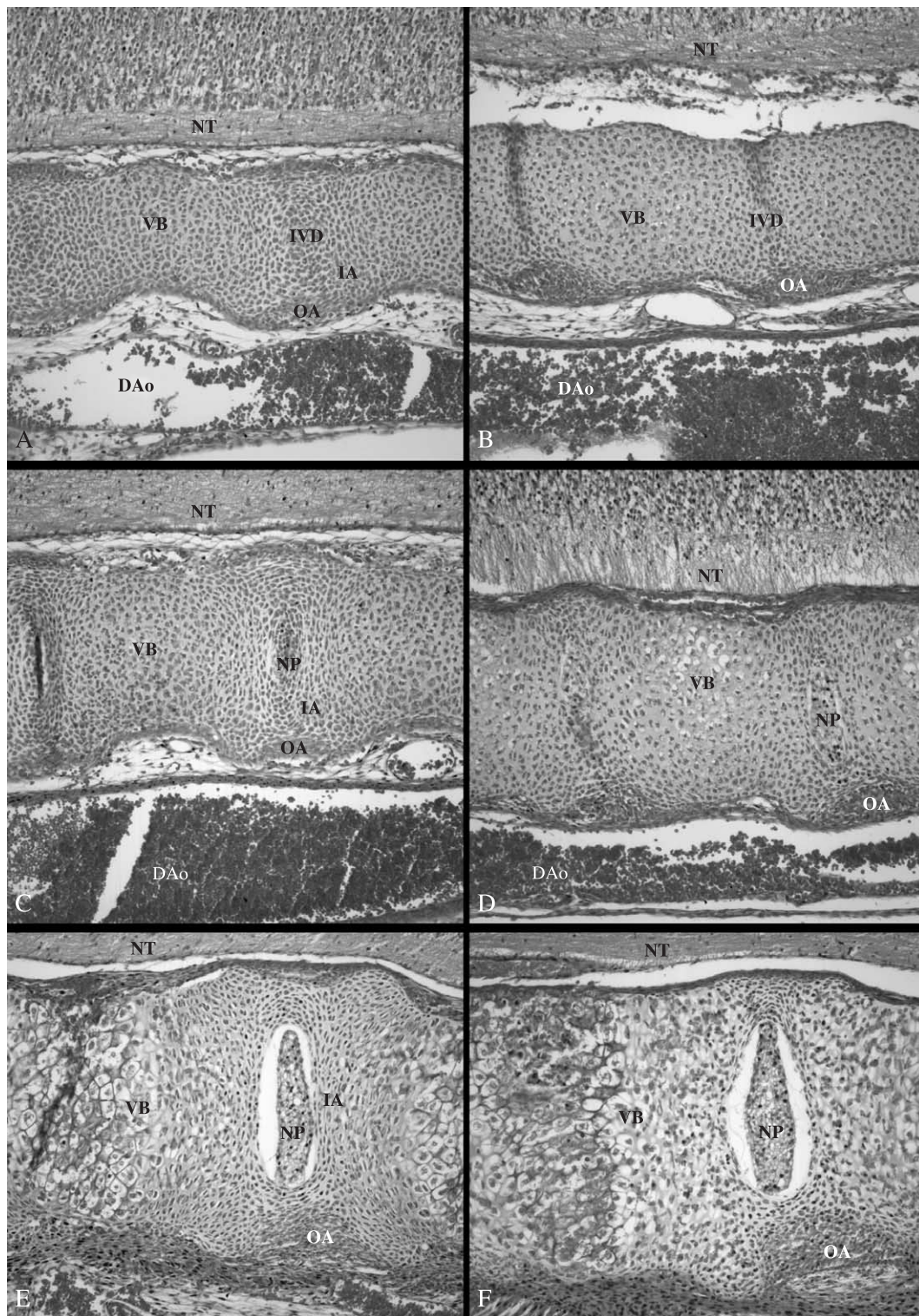


Fig. 6. Cartilage histology, intervertebral disc. Sagittal sections showing upper thoracic vertebrae at the level of the lungs and heart, E15.5 days (A–D). Lateral sections are shown in A and B; medial sections are shown in C and D. Vertebral bodies (VB), IVD, nucleus pulposus (NP), outer annulus (OA), and inner annulus (IA) are clearly demarcated in the sections from control mice (A, C). The regions of the vertebral body and the IVD are not clearly demarcated in the experimental mice (B, D). Specifically, the inner annulus is not well defined. Sagittal sections showing midthoracic vertebrae at the level of the liver, E17.5 days (E, F). Medial sections showing the vertebral body (VB), the nucleus pulposus (NP), and inner and outer annulus (IA, OA) are shown. Note that the disorganization of the IA persists at 17.5 days of gestation in the experimental mice. The neural tube (NT) is shown for reference.

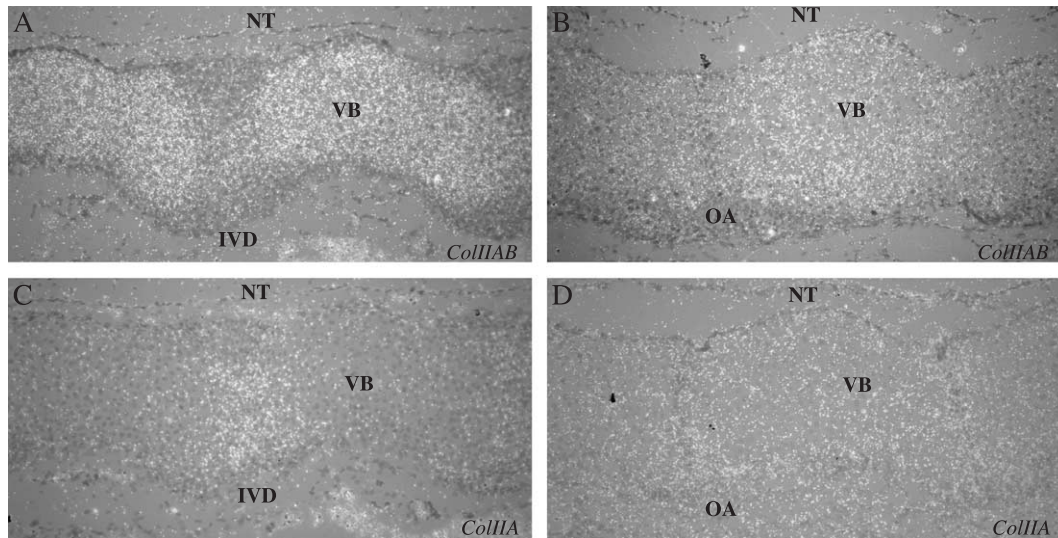


Fig. 7. In situ hybridization for markers of cartilage differentiation, IVD. Sagittal sections from control (A, C) and experimental (B, D) embryos at 15.5 days of gestation were hybridized to antisense riboprobes directed to overlapping sequences of types IIA and IIB collagen (A, B; ColIIAB) or to unique sequences in collagen type IIA (C, D; ColIIA). Dark field and toluidine blue-stained bright field images were merged with Photoshop software. Hybridization is seen as bright white grains on the gray background. The areas of the future vertebral body (VB) and intervertebral disc (IVD) are indicated. The outer annulus is indicated on sections from Col2acre^{+/-};Tgfb2^{loxP/loxP} mice. The neural tube (NT) is marked for reference. The segmentation between VB and IVD is clearly demarcated in control mice by stripes of ColIIA expression. Segmentation is not clear in Col2acre^{+/-};Tgfb2^{loxP/loxP} mice.

expressed in all prevertebral segments with the highest levels of expression in the thoracic and lumbar sclerotome. TGF- β 3 mRNA is also expressed in all prevertebral segments later becoming restricted to the perichondrium and intervertebral discs. The effect of exogenous TGF- β on vertebral development has not been studied; however, the effects of other members of the TGF- β superfamily on development of the vertebrae have been addressed. Mis-expression of CDMP-1, the human homolog of GDF5, in cartilage and notochord of transgenic mice using DNA regulatory elements from the type XI collagen promoter [α 2 (XI)] resulted in alterations in the formation of both the appendicular and axial skeleton (Tsumaki et al., 1999). In the vertebrae, expression in the notochord appeared to inhibit the migration of sclerotomal cells to the notochord resulting in the absence of vertebral bodies. CDMP-1 is not normally expressed in the axial skeleton, but it is thought to share some of the same receptors as BMPs. Bmp4 and the BMP receptors, Bmpr1a and Bmpr1b, are expressed in sclerotome, dorsal ectoderm, and mesoderm (Monsoro-Burq et al., 1996; Nifuji et al., 1997; Yi et al., 2000). Exogenous addition of BMP2 or BMP4 to the dorsal portion of the developing chick vertebrae promoted the formation of extra dorsal cartilage (Monsoro-Burq et al., 1996; Nifuji et al., 1997). The results indicate that exogenous BMP signaling is sufficient to induce the formation of dorsal vertebrae, but they do not address the normal role of BMP signaling in this system or the necessity of BMPs for vertebral development.

The role and necessity of members of the TGF- β superfamily in specific aspects of skeletal development are most clearly illustrated in mice and humans with mutations or targeted deletions in their respective genes (Chang et al.,

2002; Kingsley, 1994; Serra and Chang, 2003; Zhao, 2003). Mutations in human or mouse *Cdmp1/Gdf5* result in premature closure of distal growth plates leading to short-limbed dwarfism, but the axial skeleton and facial bones are not affected. More recently, it was found that mutations in an antagonist of BMP activity, *noggin*, have been associated with two autosomal dominant skeletal disorders, proximal symphalangism and multiple synostoses syndrome (Gong et al., 1999). Multiple synostoses syndrome is characterized by the fusion of several joints in the hands and feet as well as in the hip and spine. *Bmp4*-null mice die around the time of gastrulation with no mesoderm making studies of vertebral development difficult (Winnier et al., 1995). Mice null for *Bmpr1a* also die early in gestation (Mishina et al., 1995), while mice with a deletion of *Bmpr1b* are viable and demonstrate defects restricted to the development of the appendicular skeleton (Yi et al., 2000). In contrast, targeted deletion of the *Tgfb2* gene results in several skeletal abnormalities including alterations in size and shape of limb rudiments, bifurcation of the sternum, as well as defects in the closure of the neural arches in the lumbar regions of the spine (Sanford et al., 1997). Similarly, Col2acre^{+/-};Tgfb2^{loxP/loxP} mice demonstrated defects in the closure of the neural arches as well as defects in the formation of transverse processes and zygapophyses in the lumbar vertebrae. The data suggest an important role for the TGF- β 2 ligand, signaling through the TGF-type II receptor, in development of the dorsal and lateral parts of the lumbar vertebrae.

In contrast to *Tgfb2*-null mice, vertebral defects in Col2acre⁺;Tgfb2^{loxP/loxP} mice were not restricted to the lower thoracic and lumbar vertebrae. Defects were observed along the entire length of the spine. In addition, interverte-

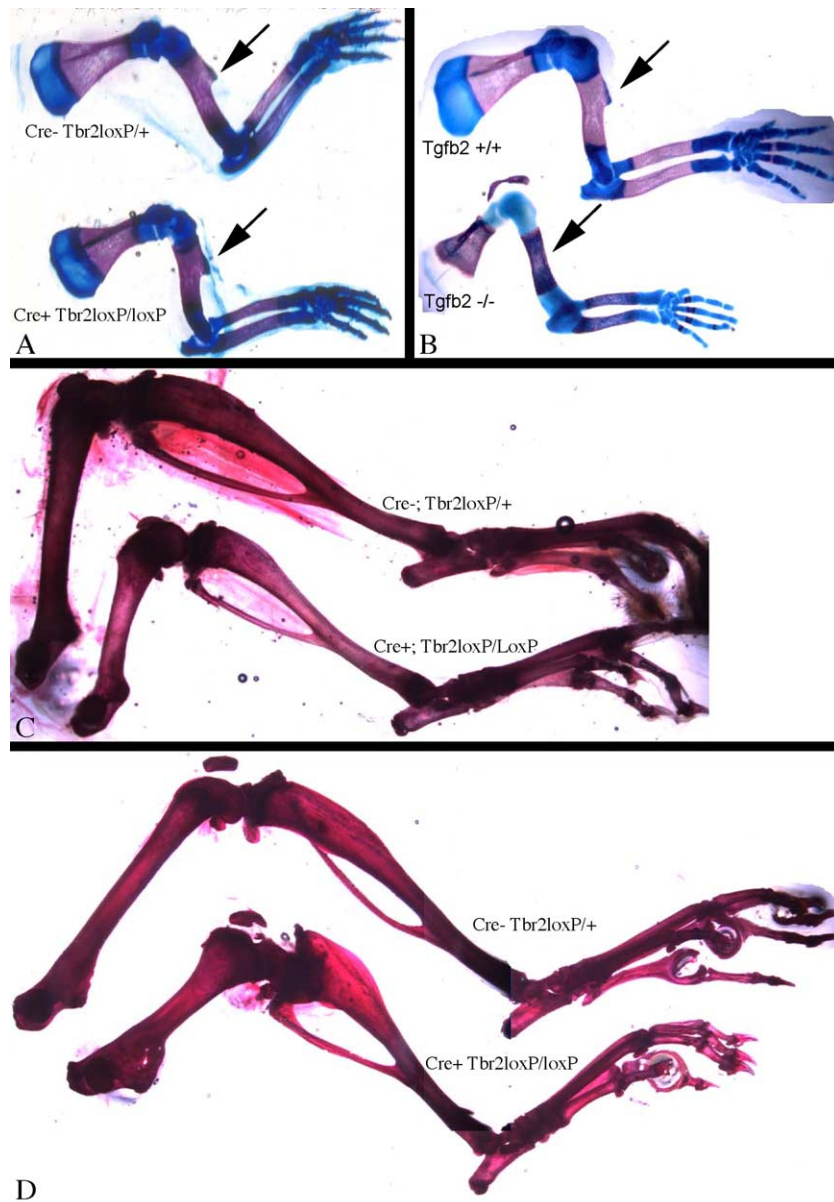


Fig. 8. Long bone development. Alcian blue-/Alizarin red-stained skeletons were prepared from E17.5 day $Col2a^{Cre-};Tgfb2^{loxP/+}$ and $Col2a^{Cre+};Tgfb2^{loxP/loxP}$ (A) as well as $Tgfb2^{+/+}$ and $Tgfb2^{-/-}$ (B) embryos. See Table 1 for morphometric data. The deltoid tuberosity is demarcated by an arrow. Alizarin red-stained skeletons were prepared from 9-week (C) and 12-week (D) Cre-negative control and Cre-positive experimental mice. Measurements of bone length are shown in Table 3.

bral discs were normal in $Tgfb2$ -null mice, while they were reduced or absent in $Col2a^{Cre+};Tgfb2^{loxP/loxP}$ mice. Furthermore, there were few similarities in the phenotype of the cranial skeleton in $Tgfb2$ -null mice and $Col2a^{Cre+};Tgfb2^{loxP/loxP}$ mice (Sanford et al., 1997). One similarity was the absence of the supraoccipital bone suggesting that the development of the supraoccipital bone is mediated by TGF- β 2 through T β RII. $Tgfb2$ -null mice demonstrated defects in most of the cranial bones that are derived through intramembranous ossification (Sanford et al., 1997). As expected, since the $Col2a$ gene is not expressed in intramembranous bones, $Col2a^{Cre+};Tgfb2^{loxP/loxP}$ mice did not present with defects in these skull bones. $Col2a^{Cre+};$

$Tgfb2^{loxP/loxP}$ mice did not have defects in the mandibles or palate that were seen in $Tgfb2$ -null mice. These structures are derived from neural crest. Conditional deletion of $Tgfb2$ in neural crest cells was previously shown to result in defects in many craniofacial bones and the palate similar to that observed in $Tgfb2$ - and $Tgfb3$ -null mice (Ito et al., 2003).

Mice with null mutations or deletions of the $Bmp7$ gene also display developmental defects in the skeleton (Jena et al., 1997; Luo et al., 1995). Defects in the basisphenoid bone overlap with those seen in $Col2a^{Cre+};Tgfb2^{loxP/loxP}$ mice raising the question of whether other members of the TGF- β superfamily can signal through T β RII. Misexpres-

Table 1

Long bone length and mineralized area, E17.5 days

Skeletal element	Genotype	Average length (mm)	Standard deviation	<i>t</i> Test <i>P</i> value	% Mineralized	Standard deviation	<i>t</i> Test <i>P</i> value
Femur	Cre ⁻ ; <i>Tgfb2</i> ^{loxP/+}	2.77	0.23	0.13	40.0	6.9	0.81
	Cre ⁺ ; <i>Tgfb2</i> ^{loxP/loxP}	2.62	0.10		39.4	0.60	
Tibia	Cre ⁻ ; <i>Tgfb2</i> ^{loxP/+}	2.87	0.21	0.02	47.4	8.5	0.78
	Cre ⁺ ; <i>Tgfb2</i> ^{loxP/loxP}	2.64	0.14		46.4	5.2	
Humerus	Cre ⁻ ; <i>Tgfb2</i> ^{loxP/+}	3.14	0.26	0.07	45.1	9.6	0.31
	Cre ⁺ ; <i>Tgfb2</i> ^{loxP/loxP}	2.96	0.07		40.9	6.1	
Radius	Cre ⁻ ; <i>Tgfb2</i> ^{loxP/+}	2.43	0.13	0.11	51.2	7.0	0.20
	Cre ⁺ ; <i>Tgfb2</i> ^{loxP/loxP}	2.26	0.24		47.8	2.7	
Femur	<i>Tgfb2</i> ^{+/+}	2.56	0.18	0.10	36.9	3.9	0.0009
	<i>Tgfb2</i> ^{-/-}	2.4	0.11		46.3	2.5	
Tibia	<i>Tgfb2</i> ^{+/+}	2.81	0.17	0.002	46.4	2.8	6.2E-5
	<i>Tgfb2</i> ^{-/-}	2.42	0.13		57.9	3.1	
Humerus	<i>Tgfb2</i> ^{+/+}	3.11	0.14	0.91	46.4	2.2	0.024
	<i>Tgfb2</i> ^{-/-}	2.93	0.18		51.8	4.3	
Radius	<i>Tgfb2</i> ^{+/+}	2.44	0.12	0.000154	52.1	3.9	0.0007
	<i>Tgfb2</i> ^{-/-}	2.00	0.14		65.4	5.4	

Cre⁻; *Tgfb2*^{loxP/+}, *n* = 10; Cre⁺; *Tgfb2*^{loxP/loxP}, *n* = 8.*Tgfb2*^{+/+}, *n* = 3; *Tgfb2*^{-/-}, *n* = 3.

sion data in vitro have suggested that TβRII can interact with type I receptors or ligands that mediate signals for BMP or activin (Ten Dijke et al., 1996). The relevance of these findings for normal in vivo development has not been demonstrated. In addition, it has been reported that a dominant-negative form of the TGF-β type II receptor (DNIIR) can block BMP activity in cell culture (Dumont and Arteaga, 2003). In contrast, others have shown that a similar DNIIR does not affect activin signaling in hepatocytes (Bottinger et al., 1997). Furthermore, the phenotypes observed in transgenic mice that express DNIIR (Serra et al., 1997) overlap with those observed in mice null for TGF-β-specific signaling genes including (1) *Tgfb2* (Sanford et al., 1997), (2) *Smad3*, a downstream mediator of TGF-β and activin signaling (Yang et al., 2001), and (3) *Ltbp3* (Dabovic et al., 2002), a binding protein specific for TGF-β. The phenotype of Col2acre⁺; *Tgfb2*^{loxP/loxP} mice does not overlap with those reported for other components of the BMP or activin signaling pathways (except for BMP7 as described above) (Chang et al., 2002; Zhao, 2003). In any case, the data indicate that *Tgfb2* is required for development of the axial skeleton, and it remains a formal possibility that

ligands other than TGF-β are signaling through this receptor.

Intervertebral discs

The IVDs are derived from the ventral sclerotome (Christ et al., 2000; Theiler, 1988). Initially, the cells form a continuous tube around the notochord. The tube then becomes segmented into the future IVD area and the vertebral body. The notochord is removed from the vertebral bodies and expands in the area of the IVD to form the nucleus pulposus. The nucleus pulposus forms the inner layer of the disc, and it is rich in proteoglycans. The outer layer, the annulus fibrosus, is divided into inner and outer annuli. The inner annulus is cartilaginous and expresses type II collagen. The outer annulus is fibrous and expresses type I collagen. Very little is known about the molecular mechanisms that govern development of the intervertebral disc. Mice with null mutations in the *Col2a1* gene have defects in the formation of the IVD (Aszodi et al., 1998). Specifically, the notochord persists in the vertebral body and does not expand into the disc area. Mice with a conditional deletion of *Jun* in Col2a expressing cells do not develop intervertebral discs because of increased apoptosis in the notochord (Behrens et al., 2003). It was also recently demonstrated that Sox5 and Sox6 are required for survival of the notochord and development of the nucleus pulposus (Smits and Lefebvre, 2003). In contrast, the nucleus pulposus in Col2acre⁺; *Tgfb2*^{loxP/loxP} develops normally, and it is removed from the vertebral body but does not expand fully into the disc area. The inner annulus does not form, and mineralization can be seen from the vertebral body into the area where the IVD cartilage should be maintained. The expression patterns of types IIA and IIB collagens suggest that *Tgfb2*

Table 2

BrdU incorporation in E17.5-day long bones

Genotype	Cre ⁻ ; <i>Tgfb2</i> ^{loxP/+}	Cre ⁺ ; <i>Tgfb2</i> ^{loxP/loxP}	<i>Tgfb2</i> ^{+/+}	<i>Tgfb2</i> ^{-/-}
% Labeled nuclei	8.60	10.98	9.05	7.22
average				
Standard deviation	1.84	2.56	2.21	0.92
<i>t</i> Test <i>P</i> value		0.097		0.035
<i>n</i>	6	6	10	8

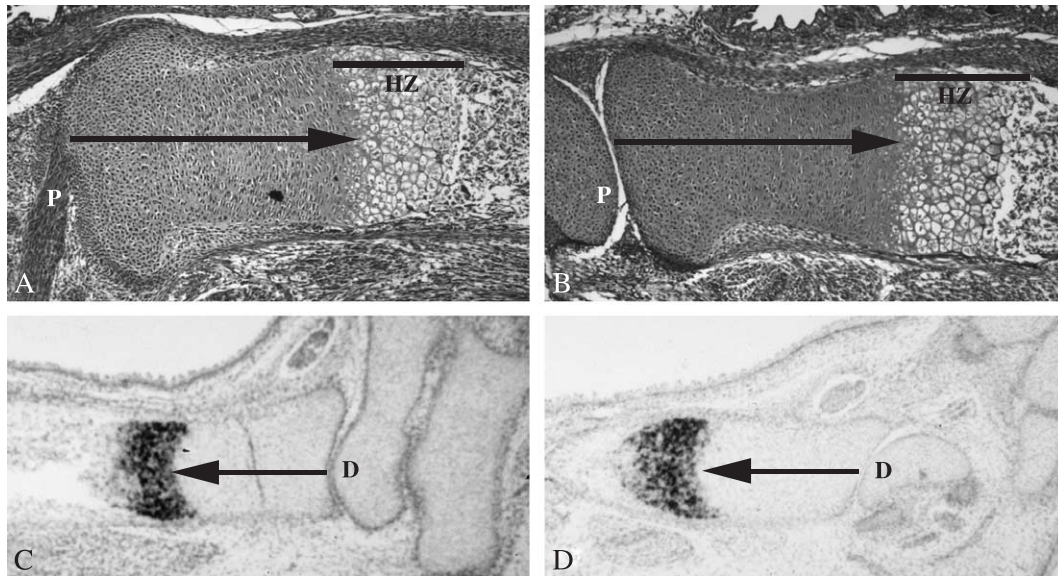


Fig. 9. Cartilage organization and differentiation in long bones. Hematoxylin and eosin-stained sections through the tibia, E17.5 days (A, B). Alterations were not detected in the organization of the proximal cartilage growth plate in control (A) versus $Col2acre^{+/-};Tgfb2^{loxP/loxP}$ (B) embryos. The hypertrophic zone is indicated (HZ). Arrows represent the length from the proximal end of the tibia to the beginning of the hypertrophic zone. Type X collagen expression in the distal tibia growth plate (C, D). Sections of tibia from control (C) and experimental (D) mice at 17.5 days of gestation were hybridized to an antisense riboprobe directed to type X collagen. Hybridization is seen as dark black grains on the toluidine blue-stained section. The arrow represents the length from the distal end of the bone to the start of the hypertrophic zone. Alterations in cartilage organization of hypertrophic differentiation were not detected in the long bones of control and $Col2acre^{+/-};Tgfb2^{loxP/loxP}$ mice.

has a role in setting up the boundaries between the vertebral body and IVD. TGF- β 3 is expressed very early in the development of the IVD and can be considered one of its earliest markers (Milani et al., 1991; Pelton et al., 1990). In adults, TGF- β 1-3 and T β R1 and T β R2 are expressed in both the nucleus pulposus and the annulus fibrosus (Matsunaga et al., 2003). Expression decreases with age. Exogenous application of TGF- β to rabbit IVD results in increased proteoglycan expression, and TGF- β is required to maintain rat IVD in organ culture (Nishida et al., 1999; Risbud et al., 2003). The studies together suggest an important role for TGF- β in the development and maintenance of IVD.

The perichondrium and chondrocytes in development of long bones

Previous findings using chick tibiotarsus cultures showed that the perichondrium can elaborate signals that negatively regulate both chondrocyte proliferation and differentiation (Di Nino et al., 2001; Long and Linsenmayer, 1998). Parathyroid hormone (PTH) was shown to prevent increased differentiation resulting from removal of the perichondrium, suggesting that PTHrP could be one of the factors from the perichondrium that regulates hypertrophic differentiation (Long and Linsenmayer, 1998). Previously, we showed that treatment with TGF- β resulted in an increase in PTHrP mRNA in the perichondrium of metatarsals grown in organ culture (Serra et al., 1999). Enzymatic removal of the perichon-

drium from embryonic metatarsal bones in organ culture and expression of a dominant-negative form of T β R2 in the perichondrium of bones in organ culture blocked both the effects of TGF- β on inhibition of proliferation and inhibition of hypertrophic differentiation (Alvarez et al., 2001, 2002). The results indicated that effects of TGF- β are critically dependent on the perichondrium. There are two possible models for the requirement of the perichondrium. The first is a linear pathway in which TGF- β 1 regulates the expression of a factor in the perichondrium that in turn acts directly on chondrocytes. The second possibility is that parallel pathways exist in which TGF- β 1 acts directly on chondrocytes, but an additional factor, from the perichondrium, acting on chondrocytes is also required for the response. We confirmed that Cre mRNA was not detected in the perichondrium around the long bones in 15.5-day embryos using radioactive in situ hybridization. The expression pattern conformed to what was previously observed (Ovchinnikov et al., 2000). Therefore, normal embryonic long bone development in $Col2acre^{+/-};Tgfb2^{loxP/loxP}$ mice suggests that TGF- β signaling directly in chondrocytes is not required for normal development of the long bones, supporting the model in which TGF- β regulates endochondral bone formation indirectly through factors synthesized in the perichondrium. The signaling cascades through the perichondrium that regulate embryonic development may not exist in postnatal development, since alterations in the size of specific bones were noted in adult mice $Cre^{+};Tgfb2^{loxP/loxP}$ relative to control littermates.

Table 3
Bone lengths (cm) in postnatal mice

Postnatal day 3			
Skeletal element	Cre ⁻ ;Tgfb β 2 ^{loxP/+}	Cre ⁺ ;Tgfb β 2 ^{loxP/loxP}	
Femur	0.3	N/A	
Tibia	0.4	0.4	
Humerus	0.4	0.45	
Radius	0.3	0.35	
Skull	0.85	0.85	
Postnatal day 7			
Skeletal element	Cre ⁻ ;Tgfb β 2 ^{loxP/+}	Cre ⁺ ;Tgfb β 2 ^{loxP/loxP}	
Femur	0.52	0.45	
Tibia	0.65	0.45	
Humerus	0.6	0.4	
Radius	0.5	0.4	
Skull	1.2	1.1	
Postnatal week 9			
Skeletal element	Cre ⁻ ;Tgfb β 2 ^{loxP/+}	Cre ⁻ ;Tgfb β 2 ^{loxP/+}	Cre ⁺ ;Tgfb β 2 ^{loxP/loxP}
Femur	1.3	1.3	0.9
Tibia	1.5	1.6	1.1
Metatarsal	0.7	0.75	0.75
Humerus	1.0	1.0	0.7
Radius	1.0	1.0	0.8
Skull	1.8	2.2	2.2
Postnatal week 12			
Skeletal element	Cre ⁻ ;Tgfb β 2 ^{loxP/+}	Cre ⁺ ;Tgfb β 2 ^{loxP/loxP}	
Femur	1.5	1.1	
Tibia	1.7	1.35	
Metatarsal	0.8	0.75	
Humerus	1.2	1.0	
Radius	1.0	1.0	
Skull	2.1	1.8	

The differences in the phenotypes observed in the axial and appendicular skeleton can be explained by differences in Col2a and Cre expression during the development of these structures. In the axial skeleton, Cre is expressed in the sclerotome before the initiation of chondrogenesis. It is expressed at a time when sclerotomal cells are expanding, migrating, and patterning the future axial skeleton. Chondrogenesis occurs in the sclerotome subsequent to these steps. In the appendicular skeleton, Cre is expressed after condensation of the mesenchyme as the cells differentiate into chondrocytes. Chondrogenesis is not disrupted in either the vertebrae or the long bones of Cre⁺;Tgfb β 2^{loxP/loxP} mice. The exact mechanism of the phenotype observed in the axial skeleton remains to be elucidated but likely involves early effects on the sclerotome.

The results presented here indicate that signaling through the TGF- β receptor is an important component of development in the axial skeleton and thus provide information regarding mechanisms of development in these structures. The differences observed between Tgfb β 2-null and Col2a⁺;Tgfb β 2^{loxP/loxP} mice allow distinctions to be made between the roles of various isoforms

of TGF- β and between the specific cell types that are involved in the development of a skeletal element. An overlapping phenotype, like that seen in the supraoccipital bone, suggests that TGF- β 2 through T β RII mediates the development of that structure. A distinct phenotype in Col2a⁺;Tgfb β 2^{loxP/loxP} mice, for example, in the IVD, suggests that ligands other than TGF- β 2 or a combination of ligands is required. A distinct phenotype in Tgfb β 2-null mice, for example, in the long bones, suggests that signaling by TGF- β 2 in cells that do not express Col2a is involved in acquisition of the phenotype. A clear understanding of the factors involved in specific aspects of skeletal development provides a basis for detailed studies into the molecular and cellular basis of their actions.

Acknowledgments

This work was funded by Grant No. R01 AR45605 from the National Institute of Arthritis, Musculoskeletal, and Skin Diseases of the National Institutes of Health. We

thank Dr. Tom Doetschman (University of Cincinnati) for kindly providing mice with the targeted deletion of the *Tgfb2* gene.

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